Unit: Unit 3: Functions
Topic: Definition of Relation and Function, Domain, Range, and Notations

Name: ________________
Date: ___________

Challenge (Habanero)

Q1. In a gaming leaderboard, the relation between players' usernames and their scores is represented by the set of ordered pairs: (Alice, 100), (Bob, 80), (Charlie, 100). Is this relation a function?

(A) Yes
(B) No
(C) Maybe
(D) Insufficient data
(E) None of the above

Q2. What is the domain of the function $f(x) = \frac{1}{x}$ in the context of a game's data usage, where x represents the amount of data used in GB?

(A) $(-\infty, 0) \cup (0, \infty)$
(B) $(-\infty, 0)$
(C) $(0, \infty)$
(D) $[0, \infty)$
(E) $(-\infty, \infty)$

Q3. A coder uses the function $g(x) = x^2$ to calculate the score of a player in a game, where x represents the number of levels completed. What is the range of $g(x)$?

(A) $[0, \infty)$
(B) $(-\infty, 0)$
(C) $(-\infty, \infty)$
(D) $(0, \infty)$
(E) $(-1, 1)$

Q4. In a coding challenge, the relation between the number of lines of code and the time taken to execute is represented by the graph of $y = \frac{1}{x}$. Is this relation a function?

(A) Yes
(B) No
(C) Maybe
(D) Insufficient data
(E) None of the above

Q5. A gamer uses the function $h(x) = 2x + 1$ to calculate their score, where x represents the number of hours played. What is the value of $h(5)$?

(A) 11
(B) 10
(C) 12
(D) 9
(E) 8

Q6. The domain of the function $f(x) = \sqrt{x}$ in the context of a game's data usage is

(A) $(-\infty, 0)$
(B) $[0, \infty)$
(C) $(-\infty, \infty)$
(D) $(0, \infty)$
(E) $(-1, 1)$

Q7. In a tech company, the relation between employees' IDs and their salaries is represented by the table: (101, 50000), (102, 60000), (103, 50000). Is this relation a function?

(A) Yes
(B) No
(C) Maybe
(D) Insufficient data
(E) None of the above

Q8. The range of the function $g(x) = x^2 - 1$ is

(A) $(-\infty, -1)$
(B) $[-1, \infty)$
(C) $(-\infty, \infty)$
(D) $[0, \infty)$
(E) $[1, \infty)$

Open Ended Questions (FRQ)

Q9. A gaming company uses the function $f(x) = 3x + 2$ to calculate the score of a player, where x represents the number of levels completed. Determine the domain and range of $f(x)$ and explain why this relation is a function.

Q10. A coder is developing a game that uses the function $h(x) = \frac{1}{x+1}$ to calculate the player's progress, where x represents the number of hours played. Evaluate $h(2)$ and explain the meaning of this value in the context of the game.

Chef's Tips

- Encourage students to use proper function notation
- Remind students to check for any restrictions on the domain
- Emphasize the importance of understanding the context of the problem